

Title: Customer Segmentation for Malls Using Machine Learning

Business Problem

Modern shopping malls face increasing competition from online retailers and other entertainment destinations. To remain competitive and provide personalized services, malls must understand the behavioral patterns of their customers. This project addresses the need for targeted marketing strategies by segmenting customers into meaningful groups based on demographic and spending characteristics. Effective segmentation enables mall management to tailor promotions, improve customer experience, and boost sales.

Background/History

Customer segmentation has long been used in retail and marketing to better understand and serve different groups of customers. While traditional methods relied on demographic or geographic data, modern data science techniques allow for deeper behavioral segmentation.

This project analyzes mall customer data using both unsupervised (KMeans clustering) and supervised (Random Forest classification) approaches. KMeans helps uncover natural groupings based on features like income, age, and spending score, while the classifier enables future prediction of customer segments. This dual approach supports more targeted marketing strategies and personalized engagement.

Data Explanation

Dataset Description and Data Preparation

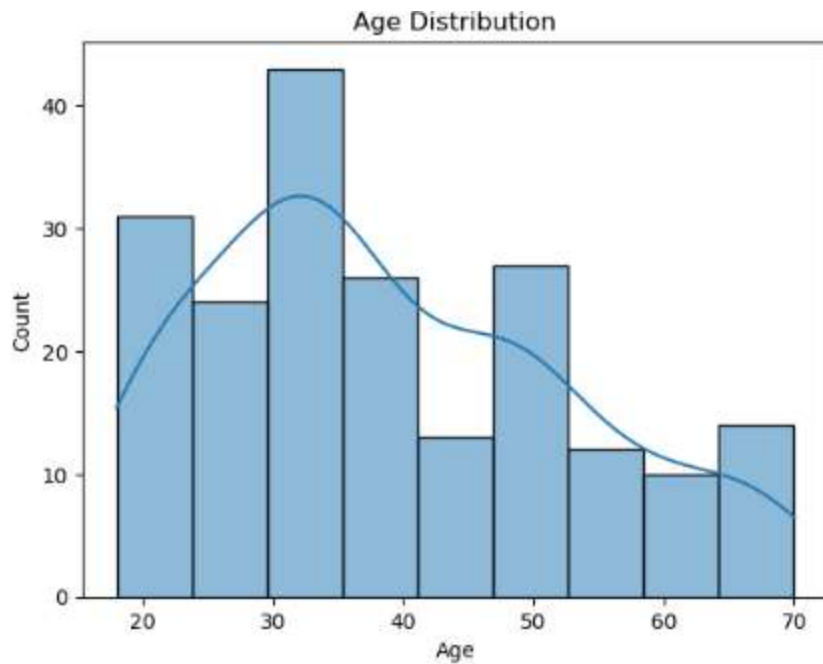
This project utilized the [Mall Customer Segmentation Data](#), a publicly available dataset hosted on Kaggle. The dataset includes detailed customer information collected by a mall, with each entry comprising key demographic and behavioral attributes such as gender, age, annual income (in thousands of dollars), and a spending score assigned by the mall based on purchasing behavior. These features are suitable for both unsupervised learning (e.g., KMeans clustering to identify segments) and supervised learning (e.g., classification models to predict segment membership). The dataset provides a strong foundation for analyzing customer patterns and behaviors.

Before analysis, the data was carefully prepared, including removal of outliers and validation of data types, ensuring a clean dataset suitable for accurate modeling and segmentation.

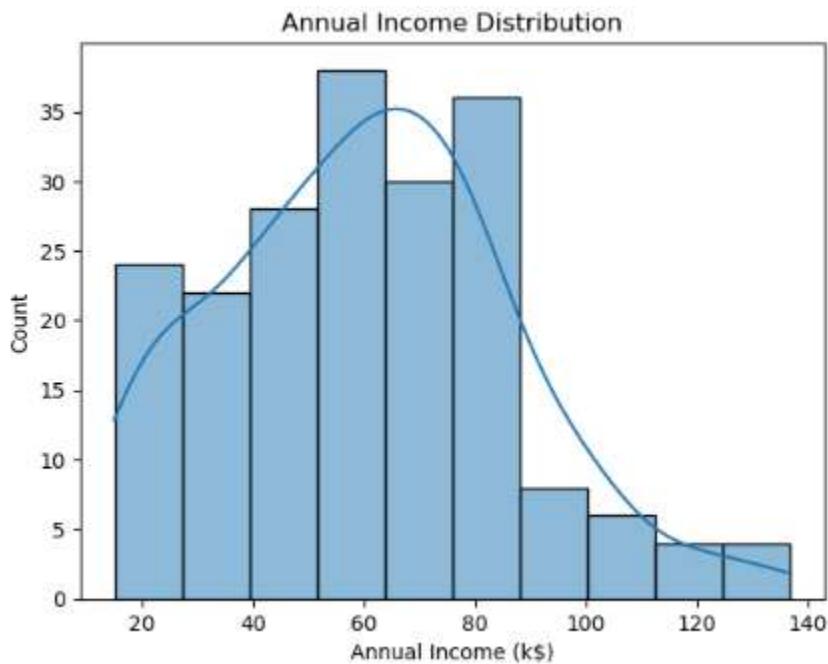
Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the distribution and variability of key features prior to modeling. This step is crucial to identify trends, detect anomalies, and guide feature engineering.

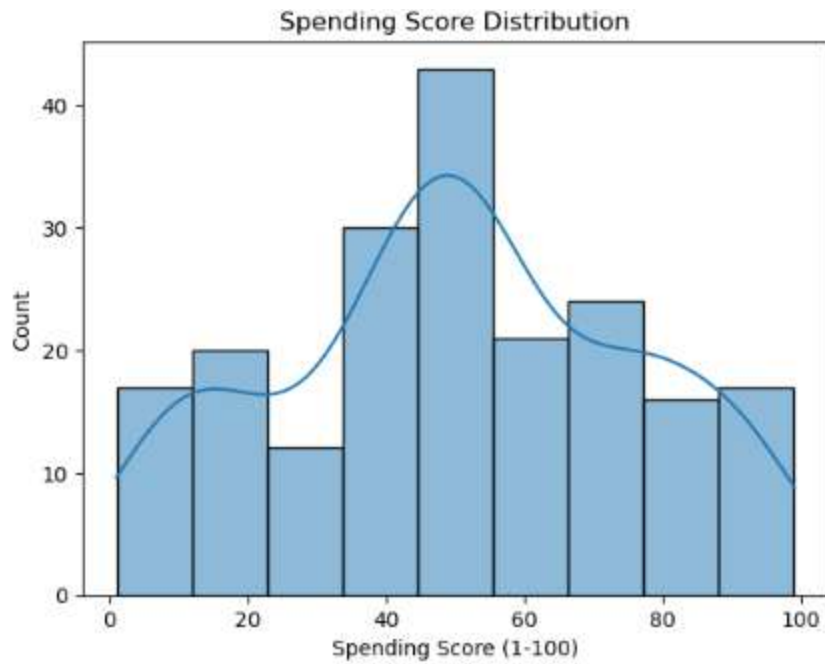
- **Age Distribution:** Most customers fall between the late 20s and mid-30s, making younger adults the core demographic.



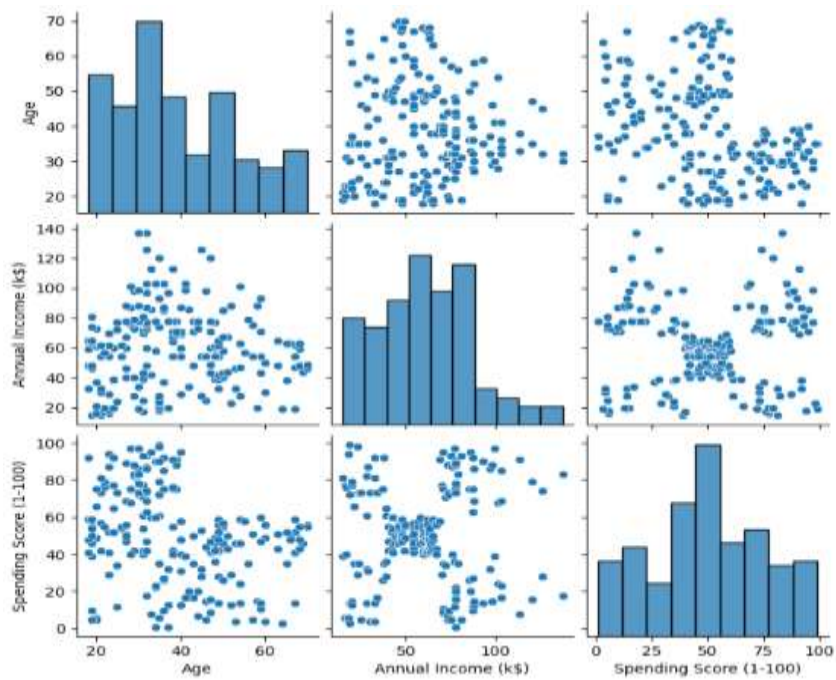
- **Annual Income Distribution:** Income is centered around 80k, with a balanced spread and few extreme values.



- **Spending Score Distribution:** Displays variability across the range, with concentrations at moderate and high spending levels.



- **Pairwise Relationships:** Pairplot visualization revealed natural groupings among income and spending behavior, showing the potential for effective clustering.



Methods and Analysis

This project used a two-phase machine learning approach. First, unsupervised clustering was used to segment customers based on spending behavior. Then, a supervised classification model was developed to predict these segments for new customers.

Phase 1: Unsupervised Learning (Clustering and Segmentation)

To identify distinct customer segments, K-Means clustering was selected for its simplicity and effectiveness. The number of clusters (k) was chosen using the Elbow Method, which identified $k = 5$ as optimal based on within-cluster variance (inertia).

Each customer was assigned to one of five clusters. Based on average feature values, interpreted the clusters as follows:

- **Cluster 0:** Low Income, High Spending
- **Cluster 1:** High Income, Low Spending
- **Cluster 2:** Young Average Spenders
- **Cluster 3:** Middle Income, High Spending
- **Cluster 4:** Older, Low Spending

Phase 2: Supervised Learning (Predicting Customer Segments)

To extend the model's practical application, a Random Forest Classifier was trained to predict a customer's segment based on the same three features: Age, Annual Income, and Spending Score. This approach allows businesses to classify new or future customers into one of the previously identified behavioral groups.

The Random Forest model was chosen due to its robustness, ability to model non-linear patterns, and low risk of overfitting with minimal tuning.

Model Evaluation and Visual Analysis

The classifier achieved 100% accuracy on the test dataset, with a perfect classification report—each segment showed precision, recall, and F1-scores of 1.00. A confusion matrix further confirmed this, revealing no misclassifications across any segment.

While these results are encouraging, such near-perfect performance should be interpreted cautiously. The dataset is relatively small, and the test set may not fully represent the broader population. There's a possibility of overfitting, especially since the classifier may have learned patterns specific to the limited data rather than generalizable trends.

To better understand model behavior, a feature importance analysis was performed. It highlighted that Annual Income and Spending Score were the most influential features in predicting customer segments, with Age contributing to a lesser degree. This finding aligns well with earlier observations from the clustering phase and the pairplot visualizations.

Overall, the combination of unsupervised clustering and supervised classification allowed for both discovery and prediction of distinct customer segments. This dual approach provides a strong foundation for data-driven, personalized marketing strategies in a mall setting.

Conclusion

This project successfully applied both unsupervised and supervised machine learning techniques to segment mall customers based on demographic and behavioral attributes. K-Means clustering uncovered five distinct customer groups, revealing meaningful patterns in income, spending

behavior, and age. Building on this, a Random Forest classifier accurately predicted segment membership, demonstrating the potential for automated customer classification in real-world scenarios. While the results are promising, further validation with larger datasets is recommended to ensure generalizability. Overall, this approach provides valuable insights to support more personalized marketing and improve customer engagement strategies.

Assumptions

- The dataset represents the mall's typical customer base and behaviors.
- Age, annual income, and spending score capture key segmentation factors.
- Outlier removal via IQR removes extreme values without losing important data.
- Standardizing features ensures balanced clustering without bias.
- K-Means clusters reflect meaningful customer groups.
- The supervised model's test performance indicates its ability to generalize despite a small sample size.

Limitations

- The dataset's size limits the generalizability of the findings.
- Spending Score is a subjective metric created by the mall and not a direct measure of monetary value.
- No temporal or transactional data to capture changing customer behavior over time.
- Limited feature diversity may omit other important behavioral factors.
- The supervised model's perfect accuracy may reflect overfitting due to limited test data.

Challenges

- Handling a small dataset limited the complexity and robustness of the models.
- Determining the right number of clusters required balancing statistical validation and interpretability.
- Avoiding overfitting in the supervised model was difficult given the limited data.
- Interpreting and labeling clusters in a way that is actionable for business users required thoughtful analysis.

Future Uses and Recommendations

This project demonstrated the effectiveness of combining unsupervised clustering (K-Means) to identify customer segments with supervised classification (Random Forest) to predict these segments for new customers. This two-step approach can be scaled to larger, more diverse datasets by adding richer features such as purchase history or online behavior. Integration with customer management systems would enable real-time, automated segmentation and personalized marketing. Periodic retraining and monitoring of both clustering and classification models will help maintain accuracy and adapt to changing customer behaviors.

Implementation Plan:

The implementation should include pipelines for preprocessing and outlier removal, followed by applying the K-Means clustering model to segment existing customers. Subsequently, a supervised classification model like Random Forest can be trained on these labeled segments to predict groups for new customers automatically. Visualization tools will support ongoing analysis and marketing strategy refinement. Regular updates of both clustering and classification models are recommended to keep pace with evolving data patterns.

Ethical Considerations

This project uses anonymized and non-sensitive customer data, focusing on general demographic and behavioral attributes to ensure privacy and compliance with regulations. Both unsupervised clustering and supervised classification models were developed with care to avoid reinforcing biases related to sensitive attributes like gender or age. The goal is to enhance personalized marketing without discrimination or exclusion. Transparency in model use and ongoing monitoring are essential to maintain fairness and trust in automated customer segmentation and prediction.

References

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<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>

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Appendix A – Supplementary Materials

This appendix includes:

- **Code Summary:** Python scripts used for data loading, cleaning, outlier removal, feature scaling, KMeans clustering, Elbow Method for cluster number selection, supervised classification using Random Forest, and evaluation via accuracy and confusion matrix.

- **Visualizations:**
 - Histograms of Age, Annual Income, and Spending Score
 - Scatter plot of Annual Income vs. Spending Score
 - Pairplot of Age, Annual Income, and Spending Score colored by cluster labels
 - Elbow plot for selecting number of KMeans clusters
 - Confusion matrix and feature importance plot for Random Forest classifier
- **Label Mapping:** Translation of KMeans cluster IDs into descriptive categories such as::
 - "Low Income, High Spending"
 - "High Income, Low Spending"
 - "Young Average Spenders"
 - "Middle Income, High Spending"
 - "Older, Low Spending"